Shift registers & shift counters

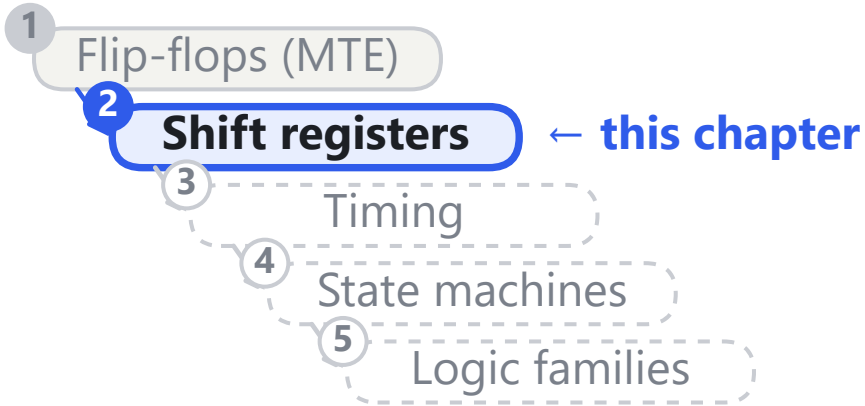
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Discuss the working of a 4 bit Johnson counter with neat block diagram.

You can't answer this yet. Six short stops from now you will, and every stop starts from the one before it.

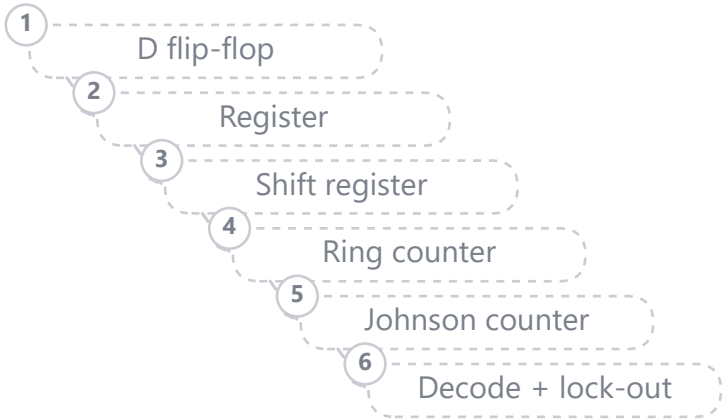
Where this chapter sits

You already have flip-flops and counters from the MTE half. This chapter takes the flip-flop one step further.



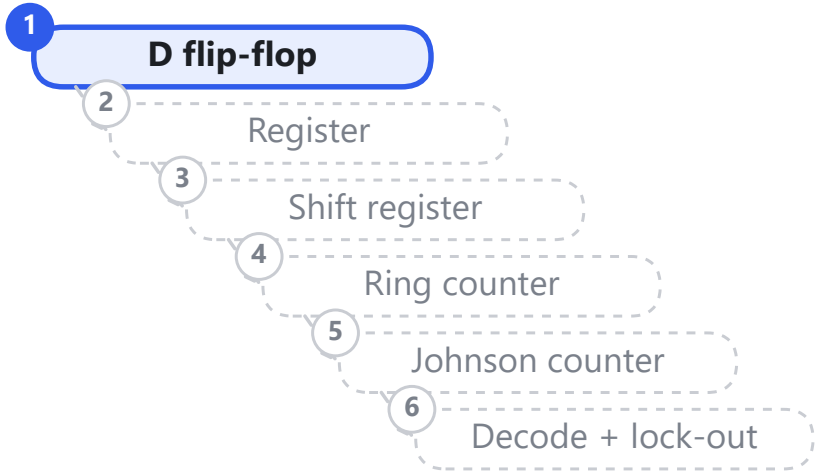
The six stops

Read top to bottom. Each stop needs only the stop above it.



STOP

D flip-flop



WORDS YOU NEED

Clock (CLK)

a signal that ticks 0 → 1 → 0 → 1 … forever

Rising edge

the instant the clock jumps from 0 to 1

D flip-flop

a 1-bit memory: input D, output Q

GUESS FIRST

Guess first

Q is 0. D is 1. The clock rises.

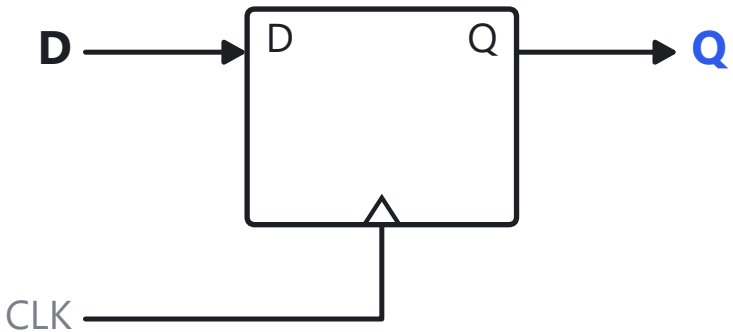
What is Q now?

Then D drops back to 0 before the next rise. **Does Q change?**

write your guess here

The rule: copy at the edge, then hold

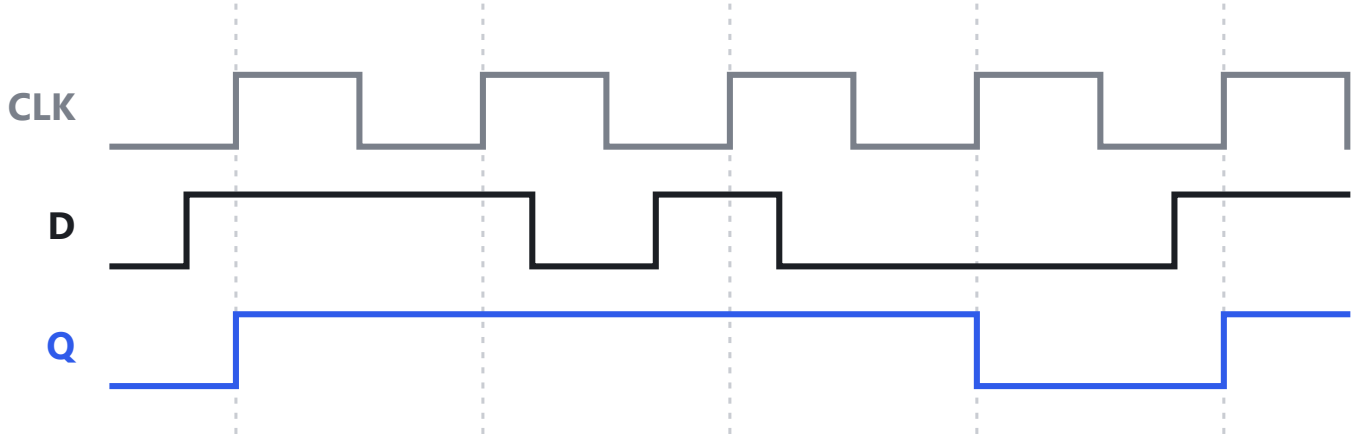
At each rising edge, Q becomes whatever D is **at that instant**. Between edges Q holds still, whatever D does.



▲ = acts only at the clock's rising edge

Watch it happen

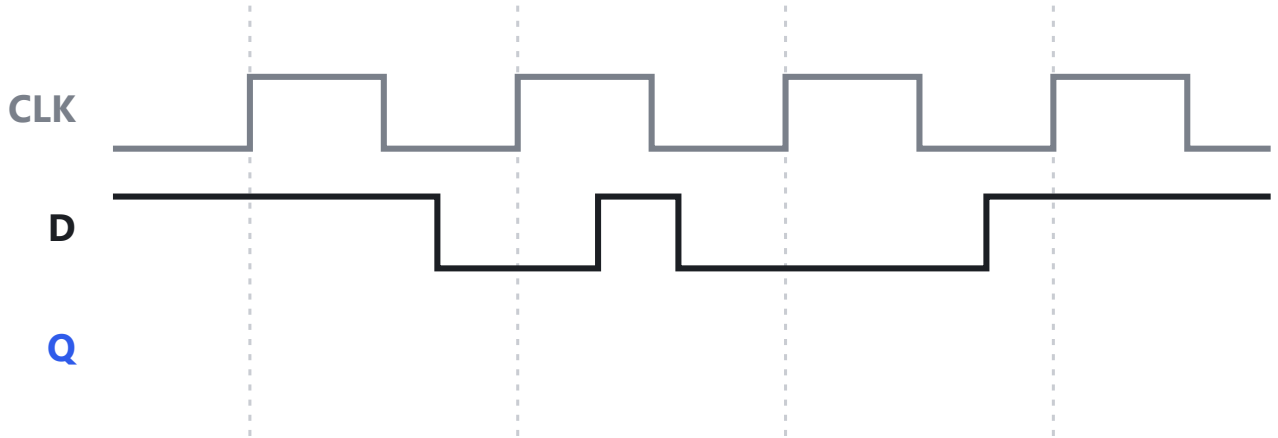
Dotted lines = rising edges. Q only moves on a dotted line. D's wiggles between edges are ignored.



CHECK

Check: draw Q

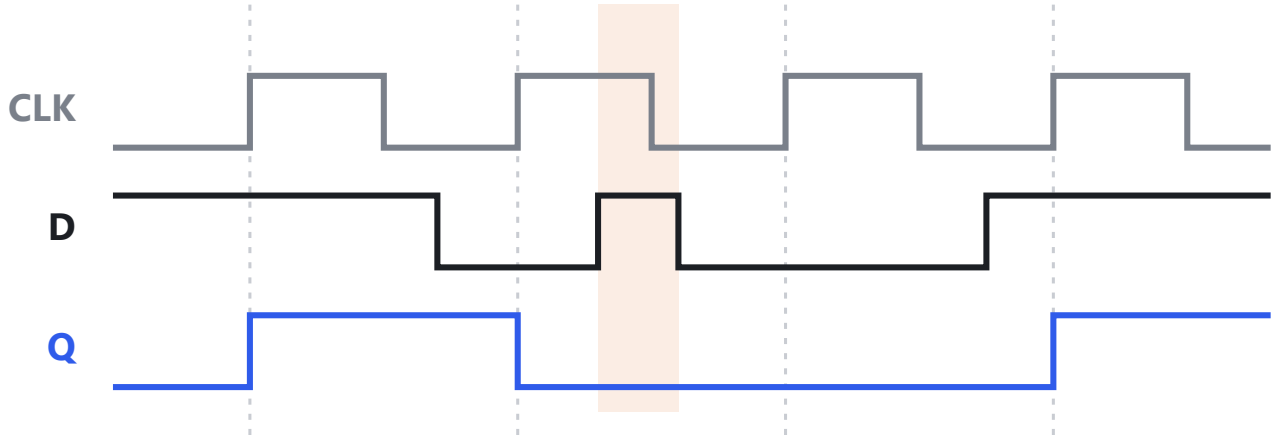
Q starts at 0. Draw Q under this D. Watch the short pulse between the 2nd and 3rd edges.



ANSWER

Answer

Q = 1, 0, 0, 1 at the four edges. The short pulse (shaded) falls **between** edges, so it is never seen.



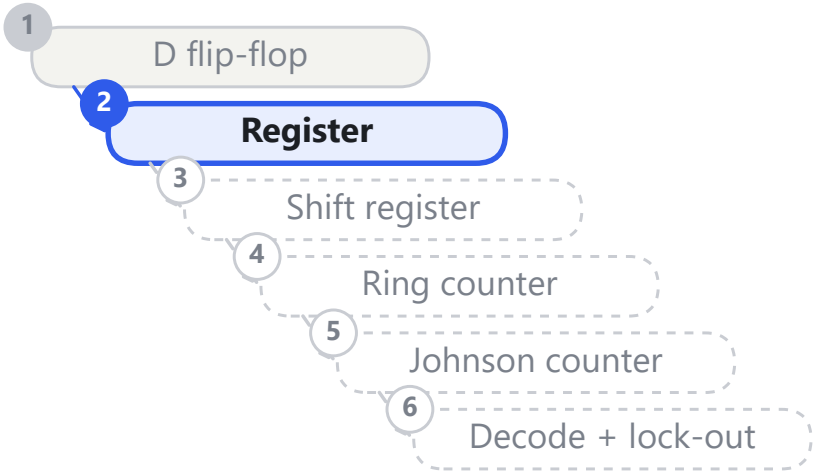
BRIDGE

One flip-flop remembers **one bit**. A 4-bit number needs four bits. So use four flip-flops.



STOP

Register



WORDS YOU NEED

Register

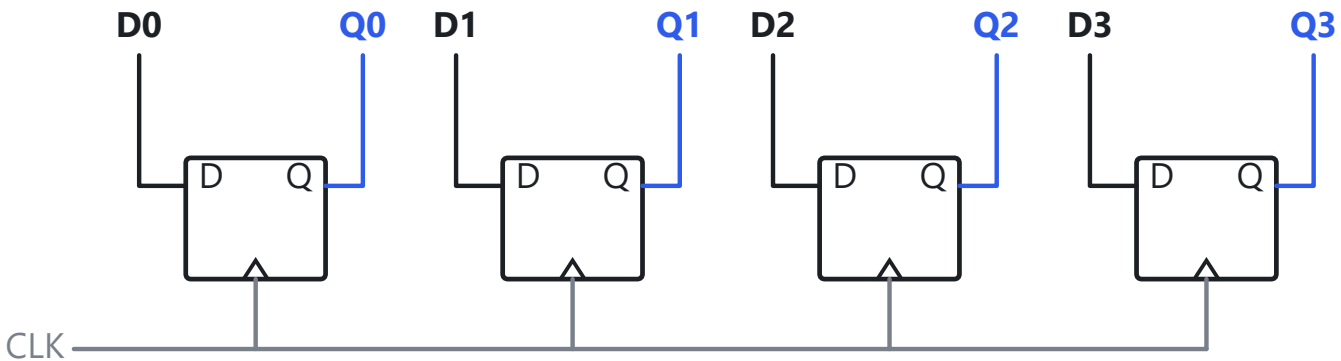
a row of flip-flops that share one clock

Parallel load

all bits enter together, at the same edge

Four flip-flops, one clock

Each flip-flop has its own D and Q. Because they share CLK, one edge copies all four bits at once. That is a 4-bit register.



one shared clock → at one edge, every Q copies its own D

CHECK

Check

D3 D2 D1 D0 = 1 0 1 1 are waiting at the inputs. Q is 0000. One rising edge arrives.

What is Q3 Q2 Q1 Q0?

write it here

ANSWER

Answer

Q3 Q2 Q1 Q0 = **1 0 1 1**. Every flip-flop copied its own D at the same edge. Nothing moved sideways.

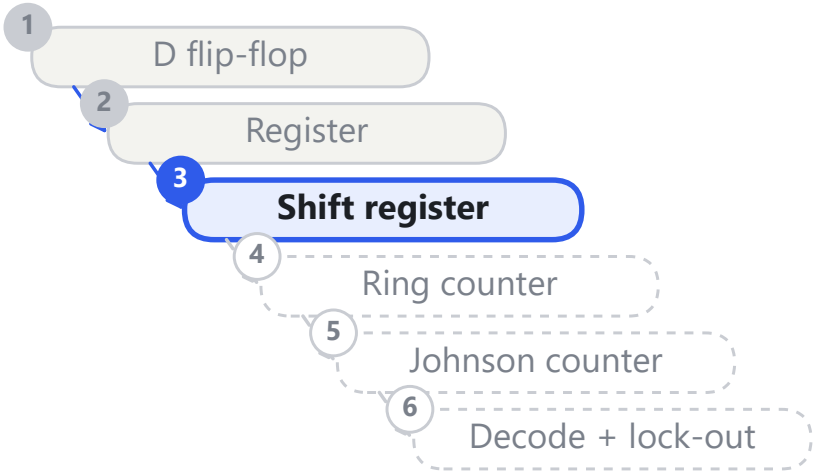
BRIDGE

In a register every input comes from outside. Now wire each flip-flop's input to its **left neighbour's output** instead.



STOP

Shift register



WORDS YOU NEED

Shift register

flip-flops in a chain: each Q feeds the next D

Serial in

the one input at the start of the chain

Serial out

the one output at the end

GUESS FIRST

Guess first

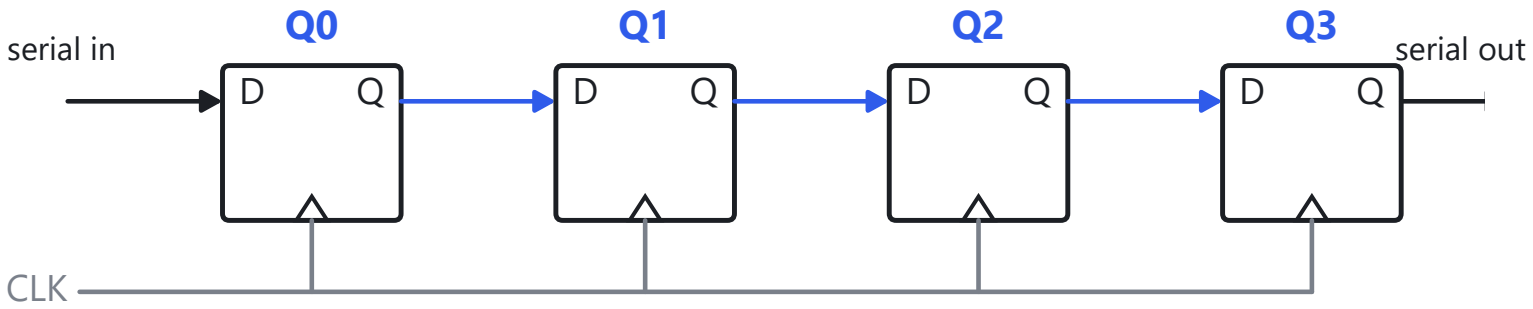
The chain holds Q0 Q1 Q2 Q3 = **1 0 0 0**. Serial in = 0. One rising edge.

What do the four flip-flops hold now?

write your guess here

Everything moves one place

Each flip-flop copies its neighbour's **old** value, because all of them sample at the same edge (stop 1). So the 1 moves one place: **0 1 0 0**.



Feeding in a word, one bit per clock

CLOCK	SERIAL IN	Q0 Q1 Q2 Q3
start	–	0 0 0 0
1	1	1 0 0 0
2	0	0 1 0 0
3	1	1 0 1 0
4	1	1 1 0 1

After 4 clocks the whole word 1011 is inside, readable all at once. Bits in one at a time, out all together: **serial-in, parallel-out**.

Why anyone builds one

USE IT AS	WHAT IT DOES
serial → parallel	a cable sends bits one at a time; the register collects a full word
parallel → serial	load a word at once, shift it out bit by bit
delay line	a bit comes out n clocks after it went in

CHECK

Check

A 4-bit chain starts at 0000. A single 1 enters on clock 1, then 0s.

On which clock does that 1 reach serial out (Q3)?

write it here

ANSWER

Answer

Clock 4. It enters Q0 on clock 1, then Q1, Q2, Q3 on clocks 2, 3, 4. A 4-stage chain delays a bit by 4 clocks.

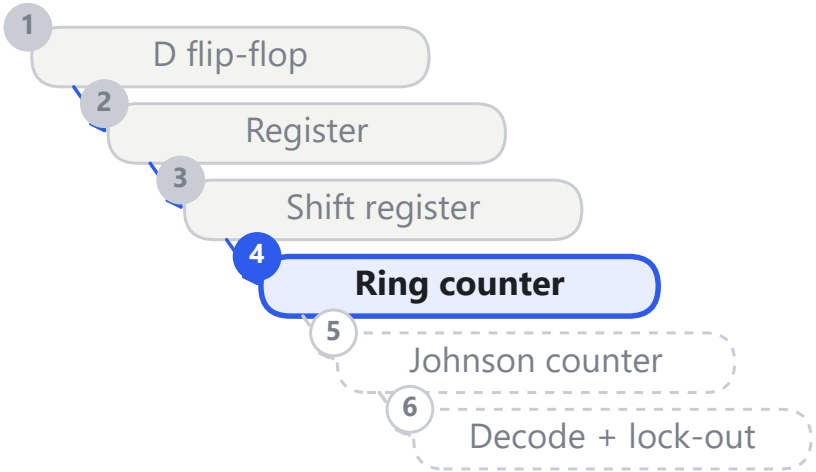
BRIDGE

In a shift register the bit that reaches the end **falls off and is lost**.
What if the end is wired back to the start?



STOP

Ring counter



WORDS YOU NEED

Counter

a circuit that steps through a fixed cycle of states, one per clock

Mod-N

the cycle has N states before it repeats

Ring counter

a shift register whose last Q feeds its first D

GUESS FIRST

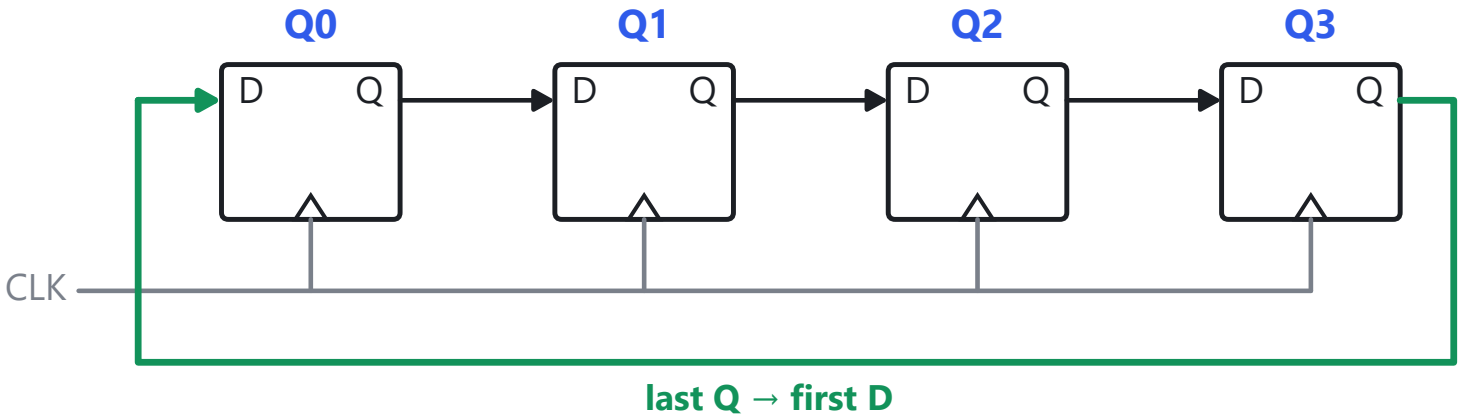
Guess first

Start at **1 0 0 0**. Q3 now feeds D0. Write the next **four** states.

write your guess here

The 1 goes round and round

1000 → 0100 → 0010 → 0001 → **1000**. Four flip-flops, four states: a **mod-4** counter. Each Q is high in exactly one state, so no extra gates are needed to read it.



One catch

It must start with exactly one 1. Start at 0000 and the zeros circulate forever: it never counts. Real designs **preset** it to 1000 at power-on.

CHECK

Check

How many flip-flops does a **mod-10** ring counter need?

write it here

ANSWER

Answer

10. A ring counter has one state per flip-flop. That's wasteful: 4 bits could hold 16 patterns, and the ring uses only 4.

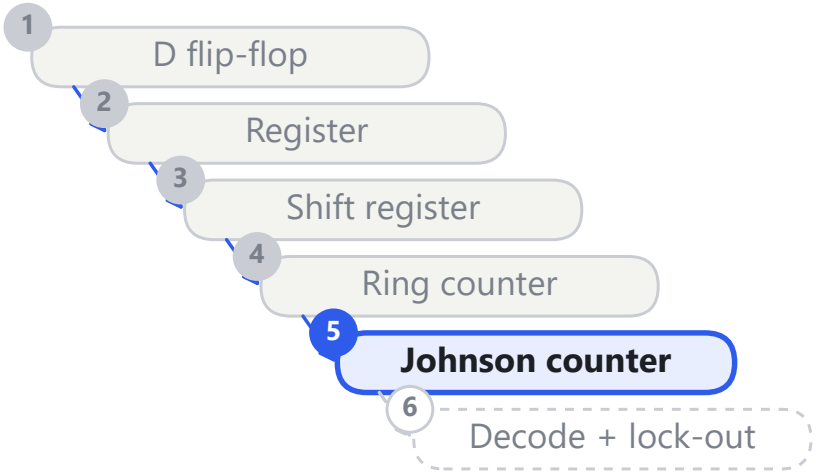
BRIDGE

The ring wastes flip-flops. **One inverter** in the feedback wire doubles the number of states.



STOP

Johnson counter



WORDS YOU NEED

Johnson counter
a ring counter that feeds back the last Q inverted (Q3′)

Q′ (Q-bar)
the opposite of Q: 1 when Q is 0

GUESS FIRST

Guess first

Start at **0 0 0 0**. D0 now gets **Q3′**. Write states until it repeats.

How many did you get?

write your guess here

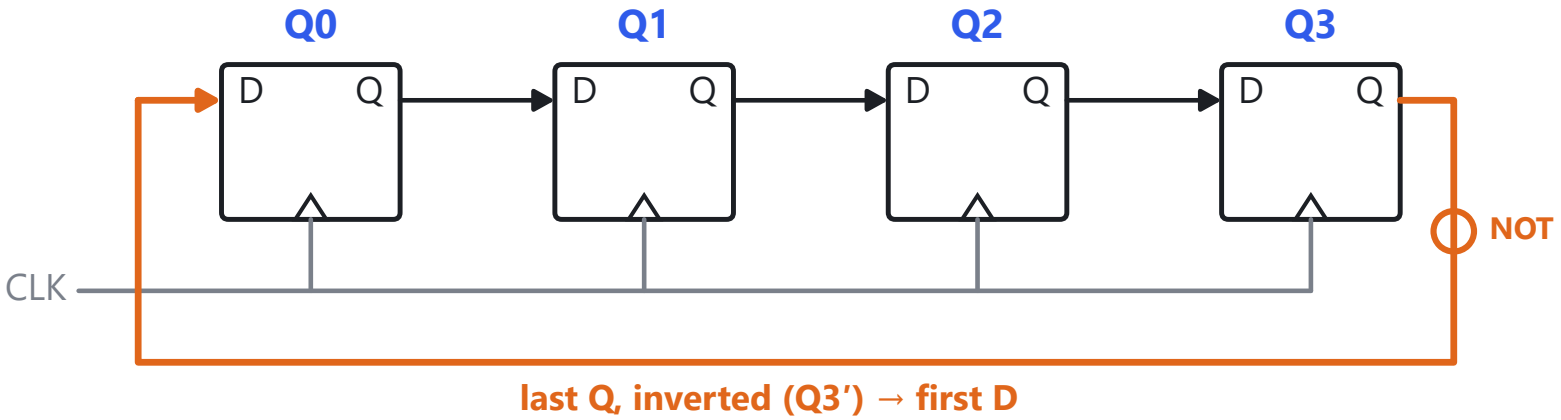
It fills with 1s, then empties

CLK	Q0 Q1 Q2 Q3	CLK	Q0 Q1 Q2 Q3
0	0 0 0 0	4	1 1 1 1
1	1 0 0 0	5	0 1 1 1
2	1 1 0 0	6	0 0 1 1
3	1 1 1 0	7	0 0 0 1

Q3 = 0 pushes in 1s; Q3 = 1 pushes in 0s. **8 states from 4 flip-flops: mod-2n.**

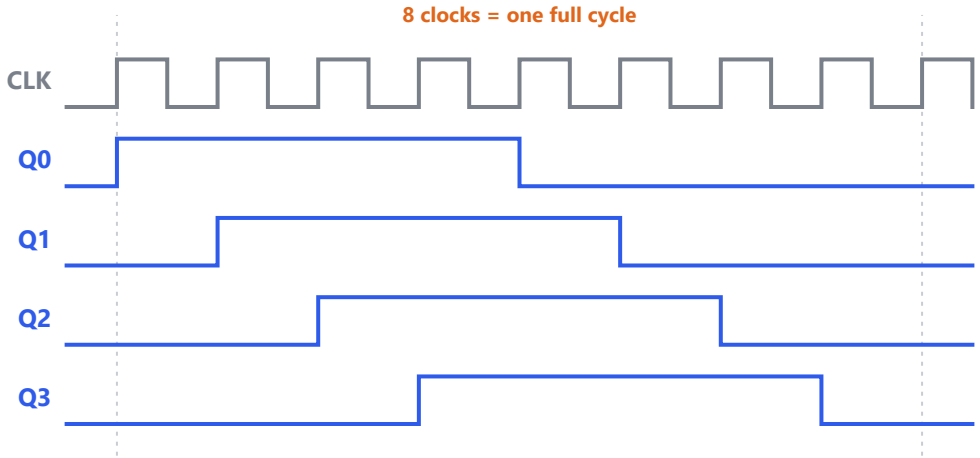
The circuit

Same chain as the ring. The only change is the inverter on the feedback wire.



The waveforms

Each Q is a square wave lasting 8 clocks: on for 4, off for 4. Each one starts one clock after the one above it.



CHECK

Check

How many flip-flops does a **mod-10 Johnson** counter need?

write it here

ANSWER

Answer

5. Johnson gives 2n states, so 2n = 10 → n = 5. The ring needed 10.

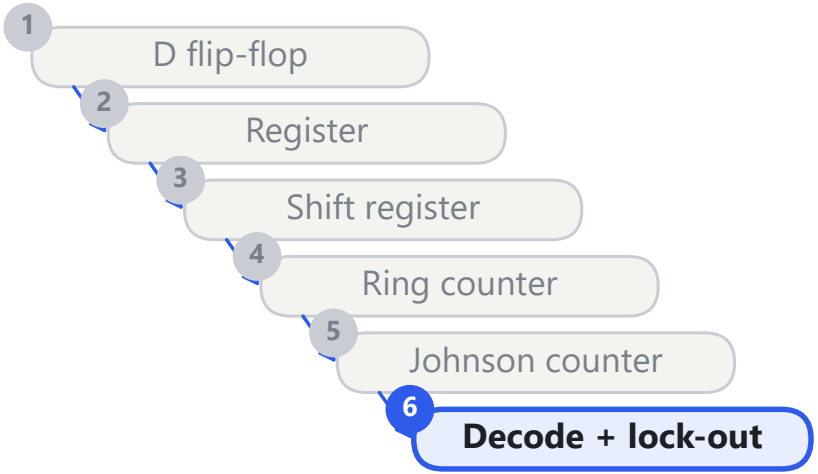
BRIDGE

A counter is only useful if some gate can **tell which state it is in**.



STOP

Decode + lock-out



WORDS YOU NEED

Decode

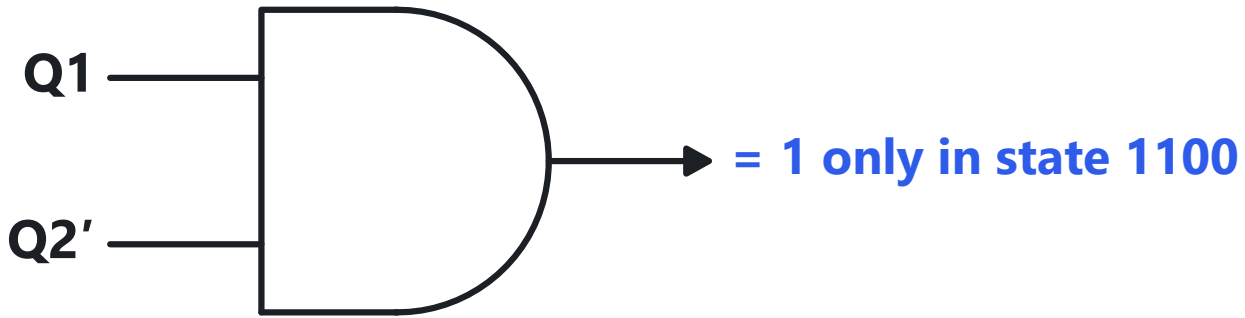
a gate whose output is 1 in exactly one state

Lock-out

the counter gets stuck cycling through states it was never meant to use

Two neighbouring bits are enough

Look at state 1100 in the table: it is the only state where Q1 = 1 and Q2 = 0. So a 2-input AND decodes it. Every Johnson state has a pair like this.



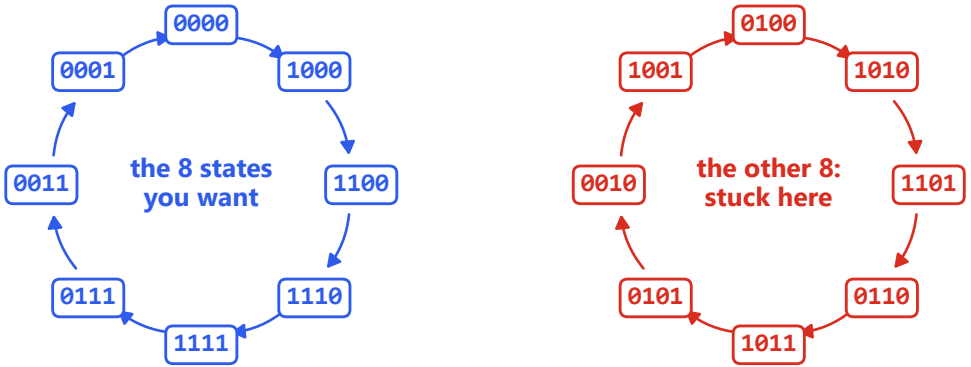
All eight decode terms

STATE	TERM	STATE	TERM
0000	Q0’Q3’	1111	Q0Q3
1000	Q0Q1’	0111	Q0’Q1
1100	Q1Q2’	0011	Q1’Q2
1110	Q2Q3’	0001	Q2’Q3

Rule: AND the two bits where 1s meet 0s.

The other 8 patterns

4 bits have 16 patterns and Johnson uses 8. The other 8 form **their own loop**. Start in one (say 0100 after a power glitch) and the counter never returns. Fix: reset to 0000 at power-on.



CHECK

Check

Which 2-input AND decodes state **0011**?

write it here

ANSWER

Answer

Q1′ · Q2. In 0011 the 0-to-1 boundary sits between Q1 (0) and Q2 (1).

Back to the exam question

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Discuss the working of a 4 bit Johnson counter with neat block diagram.

Close this PDF. Write the full 8-mark answer from memory. Use the stops as your outline. Then turn the page.

What an 8-mark answer contains

PART	FROM STOP
Definition: shift register with Q3′ fed back to D0	5
Block diagram: 4 D flip-flops, common clock, inverted feedback	3, 5
State table: 8 states, fill then empty	5
Waveforms: Q0–Q3, each f/8, one clock apart	5
Decoding with 2-input ANDs	6
mod-2n vs ring mod-n · lock-out + reset	4, 6

Chapter done: where it connects

THIS	CONNECTS TO
Ring / Johnson	synchronous counters (MTE): one shared clock, so no ripple delay
Shift register + feedback	LFSR (next): feed back an XOR of taps → pseudo-random sequence
Register	memory (Unit 5): one memory word is a register